

ELEN-0016 – Computer Vision

Student projects 2026-2027

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Introduction

Welcome to your computer vision project! Picture yourself standing on a famous square, smartphone in hand. You want to capture the perfect picture of the place, but there is a problem: tourists keep walking through your frame. Waiting for the square to be empty could take forever. What if, instead, your phone could simply make them disappear?



In this project, working in groups of three, you will build the algorithms that make this possible. Starting from videos that you will record with your own smartphone, you will reconstruct the static scene hidden behind the passers-by: first from a single viewpoint, then as a wide panorama of the whole place, captured by sweeping your phone across it, even while the crowd keeps moving.

Your mission is organized in successive phases, each one adding a new challenge on top of the previous ones:

Phase 0 – Data capture: before editing any picture, you need to take them. You will become the photographers and record your own scenes, indoors and outdoors, following a common protocol, and share them with all the other groups.

Phase 1 – Static background reconstruction: you hold your phone still while tourists walk in front of the scene. Make them vanish and recover the empty place hidden behind them.

Phase 2 – Panoramic reconstruction, moving camera, static scene: the place is finally empty, and you sweep your phone across it. Estimate how the camera moves and stitch all the frames into a single, seamless panorama of the whole place.

Phase 3 – Panoramic reconstruction, moving camera, dynamic scene: you sweep your phone across the place while the crowd keeps moving. Build the panorama while detecting and removing every passer-by, so that only the empty place remains.

Evaluation phase – The challenge: on 9 December 2026, your methods will face new scenes that none of you will have seen before.

Making the tourists disappear is only half of the fun. Once you know where they are, you can also bring them back on purpose: in Phases 1 and 3, you will insert them back onto your empty scene and panorama as a *ghost* or *trailing* effect, like an artistic long-exposure souvenir revealing how they moved through the place.

By the end of this project, you will have practiced essential computer vision techniques such as background modeling and subtraction, feature detection and matching, camera motion estimation, geometric transformations and image warping, image stitching and blending, and foreground detection and segmentation. More importantly, you will experience the full life cycle of a computer vision application: acquiring your own data, designing and implementing a method, and confronting it with data you have never seen.



Example of a panoramic view of a static scene.

Overview of the phases

Phase	Camera	Scene	Expected outputs
Phase 0 – Data capture	Static and panning smartphone	Indoor and outdoor scenes	Shared dataset of 6 videos per group
Phase 1 – Static background reconstruction	Static	Moving persons and objects	Clean background image Ghost/trailing image
Phase 2 – Panoramic reconstruction	Horizontal pan	Static (nothing moves)	Panoramic image
Phase 3 – Panoramic reconstruction	Horizontal pan	Moving persons and objects	Clean panoramic image Ghost/trailing panorama
Evaluation phase – Challenge	Static and horizontal pan	New videos recorded on-site on 9 December 2026	All the outputs of Phases 1, 2, and 3

Organization of the working teams

Each team consists of **three students**. To form your group, join the Discord of the course and send a message with your group members: you will then receive a group number, which you will use to name all your files (see Section *Schedule*).

Tools and methods

- You have to write your programs in Python 3, based on the OpenCV library or any other Python library.
- Both classical computer vision techniques and deep learning methods, including pretrained models, are allowed. However, we do not expect you to use deep learning if you consider that it is unnecessary to solve the problem: classical, unsupervised techniques can be very powerful as well. Be careful that deep learning often comes at the cost of a high computation time.
- You are allowed to reuse any algorithms available in existing libraries. We expect you to understand not only the theory behind the algorithms but also their advantages, drawbacks, and their internals.

Phase 0 – Data capture

Before any background can be reconstructed, you need a dataset. In this phase, you will record your own videos following the protocol described below, and share them with all the other groups. This shared dataset will allow every group to test their methods on a broader set of scenes and conditions than their own recordings.

Recording equipment and video format

- **Device:** all videos are recorded with your own smartphone, in landscape (horizontal) orientation.
- **Resolution:** Full HD, i.e., 1920×1080 (width $\times$ height) pixels. For performance reasons, you are allowed to process reduced versions of the images.
- **Frame rate:** at least 30 frames per second (FPS).
- **Duration:** about 1 minute per video.
- **Color:** record color videos, even if you prefer to process grayscale images.
- **Format:** MP4 files, named as described in Section *File naming and upload*.

Scene design

- For **each phase**, you must record **one indoor scene and one outdoor scene**.
- The background of the scene must be static. Prefer textured backgrounds (e.g., bookshelves, posters, building facades) and avoid plain, featureless walls, which make camera motion estimation much more difficult.
- In the dynamic scenes (Phases 1 and 3), **one to three persons** move in front of the camera. In addition to the moving persons, the scenes have to contain some other moving objects that are not human beings (e.g., balls, animals, cars). You should make the scenes as dynamic as possible.

Capture protocol

For each scene (indoor and outdoor), record the following videos.

Phase 1: static camera, dynamic scene

1. Place your smartphone on a tripod or any stable stand: the camera must remain perfectly still during the entire recording.
2. Record a video of about 1 minute, during which one to three persons, as well as some other moving objects, move in front of the scene.

Phase 2: panning camera, static scene

1. Standing at a fixed position, pan the camera horizontally from left to right, then back from right to left, for about 1 minute.
2. The only allowed movement of the camera is panning: the camera rotates horizontally around a fixed position, without tilting and without moving forward, backward, or sideways. Keep the motion smooth.
3. No moving person or object may appear in the video.

Phase 3: panning camera, dynamic scene

1. Standing at a fixed position, pan the camera in the same way as in Phase 2 (from left to right, then back from right to left), for about 1 minute.
2. During the recording, one to three persons, as well as some other moving objects, move in the scene.

File naming and upload

Each group provides the following **6 videos**, where *X* is replaced by your group number:

File names	Phase	Content
<i>group_X_phase1_static_indoor.mp4</i> <i>group_X_phase1_static_outdoor.mp4</i>	Phase 1	Static camera, moving persons and objects.
<i>group_X_phase2_pan_indoor.mp4</i> <i>group_X_phase2_pan_outdoor.mp4</i>	Phase 2	Panning camera, static scene.
<i>group_X_phase3_pan_dynamic_indoor.mp4</i> <i>group_X_phase3_pan_dynamic_outdoor.mp4</i>	Phase 3	Panning camera, moving persons and objects.

Upload your videos on Google Drive through the link provided on the Discord of the course (see Section *Schedule*).

Phase 1 – Static background reconstruction

In this first phase, you hold your phone still in front of the scene, while tourists and other moving objects keep passing in front of it. Your goal is to recover the static background as if nobody had ever been there. This phase will test your ability to model a background and to handle occlusions, people who stop for a while, and changes in lighting.

Input. A video *group_X_phase1_static_{indoor,outdoor}.mp4*, acquired with a static camera, in which people and objects move in front of the scene.

Output.

- A **clean background image** of the scene, in which all moving persons and objects have been removed and replaced by the background they were hiding.
- A **ghost/trailing image**: your clean background, onto which the moving persons and objects are inserted back at several time instants (see Section *Ghost and trailing effects*). This is your warm-up for the same effect on panoramas in Phase 3.

Phase 2 – Panoramic reconstruction: moving camera, static scene

The place is now empty, and you sweep your phone across it. Each frame only shows a part of the scene: your goal is to estimate how the camera moves from one frame to the next, and to stitch all the frames into a single, wide, and seamless panoramic view of the whole place. This phase will test your ability to estimate the camera motion, to warp images into a common coordinate system, and to blend them without visible seams.

Input. A video *group_X_phase2_pan_{indoor,outdoor}.mp4*, in which the camera pans from left to right, then back from right to left, over a static scene.

Output. A **panoramic image** of the whole area covered by the camera during the pan. Since the scene is static, no foreground removal is needed in this phase: the difficulty lies in the alignment and the blending of the frames, which should result in a panorama without visible seams, misalignments, or duplicated structures.

Phase 3 – Panoramic reconstruction: moving camera, dynamic scene

This last phase combines the two previous ones and is the most challenging. You sweep your phone across the scene while tourists and other objects keep moving in front of it. Since both the camera and the foreground are moving, a change in the image can be caused either by the camera motion or by a moving object: you will need to compensate for the camera motion and to detect the moving persons and objects, in order to remove them from the panorama.

Input. A video *group_X_phase3_pan_dynamic_{indoor,outdoor}.mp4*, in which the camera pans from left to right, then back from right to left, while people and objects move in the scene.

Output.

- A **clean panoramic image** of the whole area covered by the camera during the pan, in which all moving persons and objects have been detected and removed.
- A **ghost/trailing panorama**: your clean panorama, onto which the moving persons and objects are inserted back at several time instants, at their correct location in the panorama (see Section *Ghost and trailing effects*).

Ghost and trailing effects

Once you know where the moving persons and objects are, you can also bring them back on purpose. In Phase 1 (on the clean background) and in Phase 3 (on the clean panorama), you have to produce an image that reveals the motion of the foreground over time, directly on top of your reconstruction.

Suggested effect: fading ghosts

We suggest the following effect, in which the successive positions of each moving person or object appear as a trail of ghosts fading away into the past:

1. **Select time instants.** Choose N frames of the video, at time instants $t_1 < t_2 < \dots < t_N$, for instance regularly spaced every k frames, or selected so that successive copies of the same person do not overlap too much.
2. **Extract the foreground.** For each selected frame, compute the mask of the moving persons and objects with your foreground detection method.
3. **Align with the reconstruction.** In Phase 1, the camera is static, so the selected frames and their masks are already aligned with the clean background. In Phase 3, warp each selected frame and its mask into the coordinate system of the panorama, using your estimated camera motion, so that each copy appears at its true location in the scene.
4. **Blend the copies.** Paste the foreground pixels of the selected frames onto the clean background or panorama, from the oldest to the most recent, with an opacity that increases with time, for instance $\alpha_i = i / N$ for the i -th copy. The oldest copies then appear as faint, translucent ghosts, while the most recent one is fully opaque, which reveals the direction of motion.
5. **Refine the result.** Smooth the borders of the masks (e.g., by feathering) so that the copies blend naturally into the background.

The number of copies, their spacing, and the blending strategy are up to you, but you must be able to justify your choices.

Other effects

The fading ghosts are only a suggestion: you are free to implement any other effect of your choice, for instance a stroboscopic effect with fully opaque copies, a colored trail encoding time, or a crowd of clones of the same person spread over the panorama. Do not hesitate to be creative and take risks!

Evaluation phase – Challenge (9 December 2026)

The project ends with a challenge, held during the class on **Wednesday 9 December 2026** (time slot of the course).

- On that day, the teaching staff will record **new videos on-site**, corresponding to each phase of the project: a static camera with moving persons and objects (Phase 1), a panning camera over a static scene (Phase 2), and a panning camera with moving persons and objects (Phase 3). None of you will have seen these videos, nor these scenes, before.
- You will process these videos with your own code, on your own laptops, and produce: the clean background image and the ghost/trailing image (Phase 1), the panoramic image (Phase 2), and the clean panoramic image and the ghost/trailing panorama (Phase 3).

- Once you are done, we will check out your results and ask you questions about your design choices, parts of your code, and the values of the parameters you used.

Qualitative results only: all your results are assessed qualitatively, by visual inspection of your reconstructions and effects. No quantitative evaluation is required.

Note: do not forget that your methods should be relatively generic, as they will be applied to new scenes: avoid tuning them to your own recordings only. You are free to organize the data and the codes as you prefer (e.g., one script per phase, or a single pipeline processing all the videos). Bear in mind that you will be asked to explain some design choices, parts of the code, parameter selections, etc., so keep everything clear and easy to navigate for you for that part of the examination.

Schedule

Date	Task	Description
Deadline 25/09/2026	Discord and groups	Join the Discord and form groups of 3 students, send a message on Discord with your group members, you will receive a group number.
Deadline 07/10/2026	Phase 0	Upload your 6 recorded videos on Google Drive through the link provided on the Discord of the course.
Deadline 03/11/2026	Phase 1 checkpoint	Provide the Phase 1 outputs (clean background image and ghost/trailing image) for both your indoor and outdoor videos, and a one-page technical report following the provided LaTeX template describing your pipeline. Send your results and report by email to anthony.cioppa@uliege.be . This part is mandatory to continue the project , but is not final, as you can keep improving it until the end of the semester.

Date	Task	Description
Deadline 02/12/2026	Phases 1, 2, and 3 + Final report	Submit your code, the results of all phases on your own videos and on the videos of one of your classmates' groups, and the 2-page technical report by email to anthony.cioppa@uliege.be . You are free to continue optimizing your code until the challenge.
09/12/2026 Time slot of the course	Challenge	The challenge will be done during the class. The teaching staff will record new videos on-site, which you will process with your code. Once you are done, we will check out your results and ask you questions about design choices.

Guidelines on the usage of AI for the code and the report

You may use AI tools (such as GitHub Copilot, ChatGPT, or other large language models) to help you write code and your report. However, you must clearly indicate where and how you used them in a release note at the end of your technical report. For example: *"Lines 45–78 of panorama.py were initially generated with GitHub Copilot and then modified by [...]. The introduction of the report was drafted with ChatGPT and revised by the group."*

It is your responsibility to fully understand everything you submit. During the challenge, you will be asked to justify design choices and to explain the values of the parameters you used. It is not acceptable to answer that a parameter or design decision was chosen simply because an AI suggested it.

All written content remains your responsibility. If an AI tool produces plagiarized or uncredited material, you will be held accountable. You must always review, rewrite, and reference properly. Use AI as an assistant, for brainstorming, debugging, or drafting, but not as a substitute for your own reasoning and critical thinking.

Think of AI like a tour guide in a city you do not know: it can quickly show you around, but if you follow it blindly, you will only see what it chooses to show you. Your skill as an engineer lies not in following the guide, but in finding your own way.